ADSP 31014 Statistical Models for Data Science
Course Project Part 2

(20 points)

Objective

Define a business problem based on your Course Project Part 1 data set and provide a solution using either linear regression, logistic regression, or Poisson regression modeling. Write a (up to) 3-page report to describe the problem, the final model (including its model summary) and its performance (such as goodness-of-fit metrics and model diagnostics), key considerations (such as feature engineering, variable transformation, and model selection) that led to the final model, and any shortcoming(s) the model might have. Readers of your report should acquire a good sense of your modeling approach after reading it.

Deliverables

- Fit all your write-up onto 3 pages. No more.
- The report should be in .pdf or .docx or similar document file format.
- Show R's `summary()` print out (or Python's equivalence) of the final model. It is OK if the print out is long. It won't count towards the 3 pages.
- Other R (or Python) code is optional in the report itself.
- Submit both the report and any related reproducible code (in R or Python) to Canvas.

Some considerations

- This is not a Kaggle competition. It is not about building the most complicated unexplainable model just to fit the observed data well. Some data sets are just noisier and more difficult to predict. This by itself won't necessarily affect your project score. Focus on appropriate modeling and do what the data allows you to do. In practice, knowing what a linear model can or cannot do is also a skill.
- It is OK to switch to another data set not from Course Project Part 1. Feel free to talk to the instructor if unsure.
- Show visualization(s) if appropriate e.g. correlation matrix heat map, pair plot, fitted value plot, residual plot.

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